AtlasSAT-16U

Souss Water Observation CubeSat

Engineering Portfolio | Structural & Systems Design | Wassim Ait Oubahou
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ABOUT THIS PORTFOLIO

AtlasSAT-16U is an independent, end-to-end 16U CubeSat design project conducted to demonstrate mechanical and systems engineering competency across structural verification, CAD and manufacturing-ready drawings, and systems-level design. All analyses follow ECSS and Falcon 9 RPUG standards. No institutional affiliation.

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Page	Document	Focus
1	1A : Mission Overview	Mission concept, mass/power budgets, analysis maturity status
2	3A : Structural Assembly & Drawings	CAD assembly, configuration, manufacturing drawings
3	3B : Static Structural Analysis	FEA methodology, load cases, margins of safety

KEY RESULTS

- 9 of 24 quasi-static load-angle cases verified to date per ECSS-E-ST-32C & Falcon 9 RPUG v10
- Governing case QSL-180 ($+10$ g axial / -17 g X): $\mathrm{MoS}_{\mathrm{yield}} = +1.51$, $\mathrm{MoS}_{\mathrm{ult}} = +1.38$
- First structural mode $f_1 = 287$ Hz, $+618$% above the 40 Hz minimum requirement

AtlasSAT-16U Souss Water Observation CubeSat

Independent Research Project | Document 1A | Mission Overview | Revision C

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16U CubeSat · SSO 550 km · Falcon 9 Rideshare
VNIR multispectral · Water & agriculture monitoring
Souss-Massa, Morocco (29–31 °N, 8–10 °W)

ABOUT

AtlasSAT-16U is an independent end-to-end CubeSat design project, conducted solely to demonstrate mechanical and systems engineering competency across structure, thermal, power, orbital mechanics and ADCS systems. All analyses follow ECSS and Falcon 9 RPUG standards. No institutional affiliation.

MISSION CONCEPT

AtlasSAT-16U-SWM is a 16U CubeSat designed to deliver multispectral VNIR imagery over the Souss-Massa basin in Morocco. The mission targets irrigated agriculture monitoring, reservoir extent mapping, and seasonal water body detection through NDVI, NDRE and ET-proxy indices. Data products are designed for water resource agencies and agricultural cooperatives operating in a water-stressed semi-arid environment.

OBJECTIVES

- Map surface-water extent and reservoir area
- Monitor crop condition via NDVI / NDRE
- Derive proxy evapotranspiration indicators
- Support water abstraction policy & drought response
- Validate end-to-end design methodology.

ORBIT & ROI

Parameter	Value
Orbit type	SSO, 550 km
Inclination	≈97.5°
Period	≈95.6 min
Eclipse	≈36 min / orbit
LTDN	10:30
Revisit	≤3 days (target)
ROI centre	30.4 °N, 9.6 °W
ROI area	≈15–25 km^2
Region	Souss-Massa, Morocco

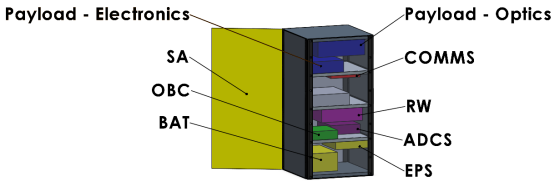
KEY SPECS

Parameter	Value
Form factor	16U (4×4×1U)
Total mass	≈24 kg (FEA sim.)
Payload	VNIR pushbroom, 5 bands
GSD at nadir	5–10 m target
Swath	≥30 km target
Data latency	≤48 h
Mission life	3 yr (goal 5 yr)
Downlink	X-band 8.025–8.4 GHz
Storage	≥512 GB

DATA PRODUCTS

- L1: Orthorectified imagery (GeoTIFF)
- L2a: NDVI / NDRE vegetation indices
- L2b: Surface-water & reservoir maps
- L2c: Evapotranspiration (ET) proxy indicators
- Users: water agencies, cooperatives, planners

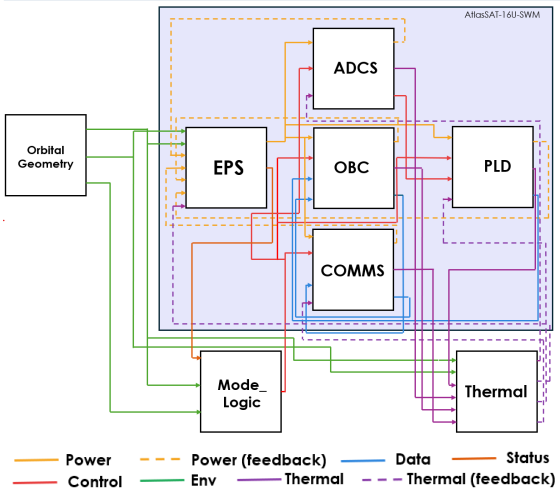
SPACECRAFT — INTERNAL LAYOUT



MASS BUDGET

Subsystem	Components	kg	%
Structure	Rails, frames, panels, plates	7.90	33
PLD optics	VNIR imager, optics, baffle	4.50	18.8
PLD electronics	Payload electronics board	1.00	4.2
ADCS	Magnetorquers	0.80	3.3
RW	Reaction wheels ×3	2.00	8.3
Battery	Li-ion pack	3.00	12.5
EPS / PCDU	Solar arrays + distribution	1.50	6.25
OBC	Computer + mass memory	0.60	2.5
COMMS	TT&C + downlink chain	0.80	3.3
Misc	Harness, brackets, misc	1.50	6.25
Total (FEA)		**≈24**	—

SYSTEM BLOCK DIAGRAM



POWER BUDGET (ESTIMATES)

Subsystem		Safe W	Nom. W	Img. W	DL W
Payload	VNIR imager	0	0	18	0
ADCS	Sensors + RW	1.5	4.0	4.0	2.0
OBC	Computer	1.5	2.5	3.0	2.5
COMMS	TT&C / DL	0.5	1.0	0	8.0
EPS losses	Regulation	1.0	2.5	3.5	3.0
Heaters	Thermal ctrl	5.0	5.0	0	0
Total (est.)		**≈10**	**≈15**	**≈28**	**≈15**
SA generation		**40 W (sunlit)**		→ 4A model	

Power values TBD pending EPS design (Doc 4B).

ENGINEERING ANALYSIS STATUS

Domain	Analysis	Status
Structural	Static FEA QSL-θ sweep (3B)	WIP
	Modal 200 modes (3C)	DONE
	Random vib. VIB-01	TBD
Thermal	Cold case SS (3D)	DONE
	Eclipse case (3D)	DONE
	Cold Transient Python (3D)	DONE
	Hot case SS (3D)	TBD
	Hot Transient Python (3D)	DONE
Orbital	Ground track / coverage (2A)	WIP
	Revisit analysis GMAT	WIP
Power	System model MATLAB (4A)	DONE
	EPS / power budget (4B)	TBD
Mission	Link budget	TBD
	ADCS jitter analysis	TBD

DONE Done *WIP* In progress *TBD* Planned

STRUCTURAL RESULTS

Check	Result	
MoS_{yield} (QSL-180)	$+1.51 > 1.25$ req.	WIP
1st eigenfreq. f_1	287 Hz $\gg$ 40 Hz	DONE
Cold-case ΔT	17 K (246–263 K)	DONE
Hot-case ΔT	96 K (260–347 K)	WIP
Eclipse-case ΔT	/	WIP
Transient cold-case ΔT	/	WIP
Transient hot-case ΔT	/	WIP

Norms: ECSS-E-ST-32C · RPUG v10

Document series: **1A** Overview · **1B** Requirements · **2A** Orbital · **3A** Assembly · **3B** Static · **3C** Modal · **3D** Thermal · **4A** System model

TOOLS USED

Tools	Use case
Solidworks	CAD Geometry Assembly
Solidworks Simulation	Static & modal FEA
Ansys Mechanical	Static & modal FEA
GMAT	Orbital propagation
VSC (Python)	Thermal transient simulation
MATLAB	Power budget modelling
Simulink	System architecture
LaTeX	Documentation

AtlasSAT-16U Souss Water Observation CubeSat

Independent Research Project | Document 3A | Structural Assembly & Drawings | Revision A

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ABOUT

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MISSION CONCEPT

AtlasSAT-16U-SWM is a 16U CubeSat designed to deliver multispectral VNIR imagery over the Souss-Massa basin in Morocco. The mission targets irrigated agriculture monitoring, reservoir extent mapping, and seasonal water body detection through NDVI, NDRE and ET-proxy indices. Data products are designed for water resource agencies and agricultural cooperatives operating in a water-stressed semi-arid environment.

OBJECTIVES

Document 3A is a portfolio overview of the AtlasSAT-16U primary structure, SolidWorks CAD assembly and representative 2D manufacturing drawings. It supports engineering demonstrations of structural design competency. Full drawings, native CAD files and a detailed BOM are held in companion report 3A-1.

CONFIGURATION

Parameter	Value
Form factor	16U ($4 \times 4 \times 1$U)
Deployer iface	P-POD / ISIPOD compliant
Rails	Al 7050-T73510 ($\times 4$)
Bulkheads	Al 6061-T6 (top, mid, bot)
Shear panels	Al 6061-T6 ($\times 4$)
Equipment plates	Al 6061-T6
Total mass	≈23 kg (FEA)
Struct. mass	7.90 kg (33%)
CoM (Y-axis)	235.7 mm
$I_{xx,yy,zz}$	0.199 / 0.641 / 0.661 kg·m^2
CAD tool	SolidWorks (ver. 2024)

SUBSYSTEM LAYOUT

Stacking order along Y axis (launch direction):

Zone	Subsystem
Upper	PLD optics (nadir-facing aperture)
Upper-mid	PLD electronics, OBC, mass memory
Centre	EPS/PCDU + Li-ion battery (CoM)
Lower-mid	ADCS: RW cluster, MTQ, IMU
Lower	COMMS: TT&C, X-band downlink

DESIGN PHILOSOPHY

- Stiffness-driven primary structure
- Clear load paths: deployer rails → bulkheads
- Conservative bonded contacts (preliminary)
- $f_1 = 287$ Hz $\gg 40$ Hz RPUG v10 minimum
- Governing case: QSL-180 ($+10$ g axial / -17 g X lateral)

CAD ASSEMBLY VIEWS

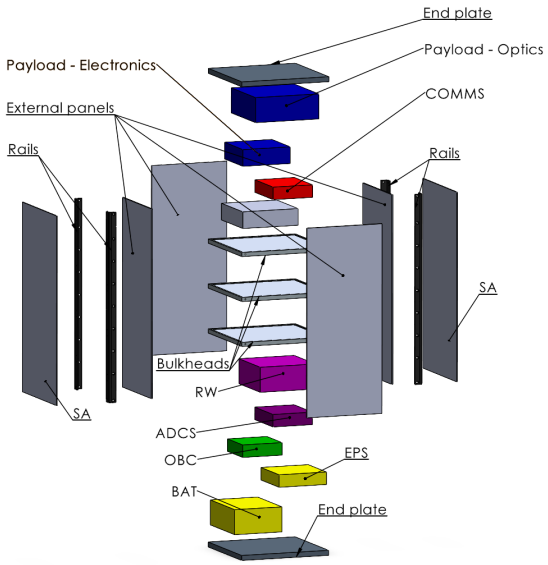


Fig. 1 — Exploded view. All structural elements and major subsystems labelled. Top-level file: `AtlasSAT16U_Assembly.SLDASM`.

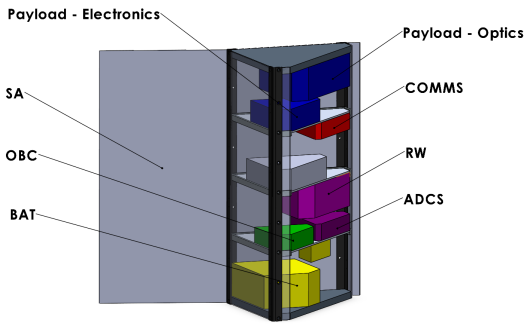


Fig. 2 — Section cut along Y (axial) axis showing internal stacking: PLD optics → OBC → EPS → ADCS → COMMS.

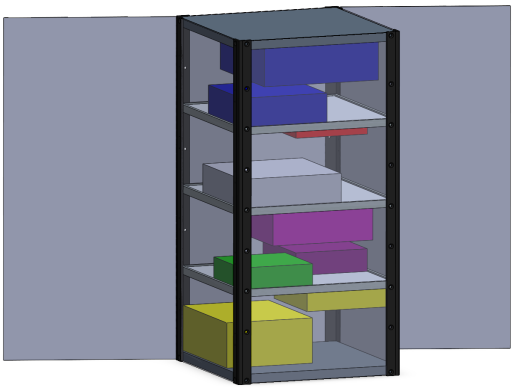


Fig. 3 — Full assembly isometric view.

ENGINEERING DRAWINGS

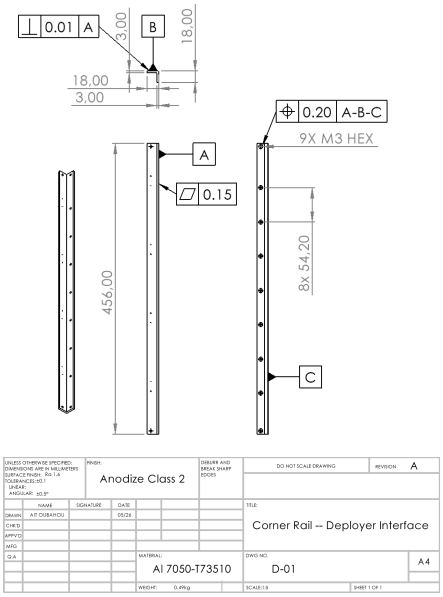


Fig. 4 — 2D drawings in SolidWorks: Corner rail D-01

Dwg	Description
D-01	Corner rail / deployer interface
D-02	Top & bottom bulkhead
D-03	Mid bulkhead (equipment deck)
D-04	Side shear panels ($\times 4$)
D-05	Payload mounting bracket
D-06	Equipment plate (ADCS/EPS zone)

MASS BUDGET

Subsystem	Components	kg	%
Structure	Rails, panels, plates	7.90	33
PLD optics	VNIR imager, optics	4.50	18.8
PLD electronics	Payload electronics board	1.00	4.2
ADCS	Magnetorquers	0.80	3.3
RW	Reaction wheels $\times 3$	2.00	8.3
Battery	Li-ion pack	3.00	12.5
EPS / PCDU	Solar arrays + distribution	1.50	6.25
OBC	Computer	0.60	2.5
COMMS	TT&C + downlink chain	0.80	3.3
Misc	Harness, brackets, misc	1.50	6.25
Total (FEA)		**≈24**	**—**

STRUCTURAL RESULTS

Check	Result	
MoS_{yield} (QSL-180)	$+1.51 > 1.25$ req.	WIP
1st eigenfreq. f_1	287 Hz $\gg$ 40 Hz	DONE
Cold-case ΔT	17 K (246–263 K)	DONE
Hot-case ΔT	96 K (260–347 K)	WIP
Eclipse-case ΔT	/	WIP
Transient cold-case ΔT	/	WIP
Transient hot-case ΔT	/	WIP

Norms: ECSS-E-ST-32C · RPUG v10

STRUCTURAL VERDICT — PASS (9/24 cases)

All 9 QSL-θ rideshare cases run to date satisfy ECSS-E-ST-32C.

$$\mathrm{MoS}_{\mathrm{yield,min}} = +1.51 > 1.25$$

$$\mathrm{MoS}_{\mathrm{ult,min}} = +1.38 > 1.50$$

Remaining 15 angles in progress; structure provisionally cleared for 3C and 3D.

AtlasSAT-16U Souss Water Observation CubeSat

Independent Research Project | Document 3B | Static Structural Analysis | Revision B
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16U CubeSat · SSO 550 km · Falcon 9 Rideshare
VNIR multispectral · Water & agriculture monitoring
Souss-Massa, Morocco (29–31 °N, 8–10 °W)

OBJECTIVE

AtlasSAT-16U is an independent end-to-end CubeSat project; this one-pager summarises the static structural (quasi-static) analysis of the primary structure, run in SolidWorks Simulation 2024. Loads follow the Falcon 9 RPUG v10 rideshare envelope (+10 g axial / +17 g lateral, swept θ = 0°–345°, 15° increment); safety factors per ECSS-E-ST-32C. Goal: demonstrate positive margins of safety at yield and ultimate across the full angular sweep. Extended mesh convergence data and full stress reports are in companion report 3B-1.

ANALYSIS MODEL

Parameter	Value
FEA tool	SolidWorks Simulation 2024
Geometry	Full assembly: L-rails, bulkheads, panels, subsystem blocks
Masses	Subsystems as density-adjusted solid blocks
Element type	Solid (curvature-based, high quality)
Mesh — global	3 mm max; 1 mm min
Jacobian	16 points
Nodes / elem.	1.3 M / 0.87 M (fasteners excluded)
Material (panels)	Al 6061-T6
Material (rails)	Al 7050-T73510
σ_y / σ_u (6061)	276 / 310 MPa
σ_y / σ_u (7050)	435 / 495 MPa
Contacts	Bonded (global, conservative)

BOUNDARY CONDITIONS

Fixed Geometry applied to the four L-rail lower end faces (face −Y, 17×17 mm per rail), representing the ISIPOD deployer contact. All 6 DOFs constrained. Fasteners excluded from simulation. Inertial loads applied via Gravity (axial, 98.1 m/s^2 at +10 g, Y) combined with Remote Load/Distributed at assembly CoM for the 17 g lateral vector, resolved into X/Z components at each sweep angle θ per the RPUG v10 rideshare envelope. Safety factors per ECSS-E-ST-32C Table 5-1.

DESIGN PHILOSOPHY

- Stiffness-driven primary structure
- Clear load paths: deployer rails → bulkheads
- Conservative bonded contacts (preliminary)
- Governing case: QSL-180 (θ=180, +10 g axial / −17 g X lateral)
- $\mathrm{FS}_{\mathrm{yield,min}} = 2.51 > 1.25$ requirement (RPUG v10)
- Structure cleared for modal and thermal phases

LOAD CASES (FALCON 9 RIDESHARE RPUG v10)

ID	θ	Axial Y	Lat. X	Lat. Z
QSL-000	0°	+10 g	+17.00 g	0 g
QSL-015	15°	+10 g	+16.42 g	+4.40 g
QSL-045	45°	+10 g	+12.02 g	+12.02 g
QSL-090	90°	+10 g	0 g	+17.00 g
QSL-135	135°	+10 g	−12.02 g	+12.02 g
QSL-180	**180°**	**+10 g**	**−17.00 g**	**0 g**
QSL-225	225°	+10 g	−12.02 g	−12.02 g
QSL-270	270°	+10 g	0 g	−17.00 g
QSL-315	315°	+10 g	+12.02 g	−12.02 g

9 of 24 angles complete (θ = 0°–345°, 15° increment). Lateral vector resolved into X/Z at each θ; magnitude 17 g throughout. Safety factors: 1.25 (yield) and 1.50 (ultimate) per ECSS-E-ST-32C.

FEA FIGURES

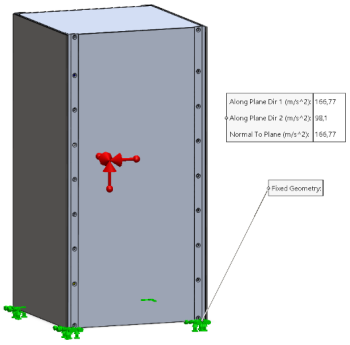


Fig. 1 — Boundary conditions: Fixed Geometry at rail base, remote load applied at assembly CoM (component magnitudes shown are illustrative of the load-definition setup).

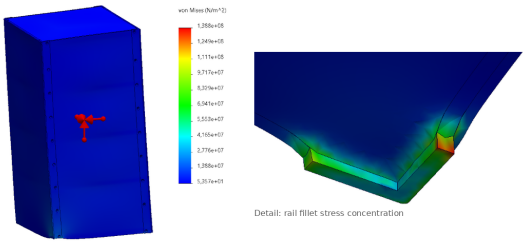


Fig. 2 — von Mises stress, QSL-180 (governing), with rail-fillet stress-concentration detail. Peak σ = 138.8 MPa at node 3405209 (X = 95.2, Y = −1, Z = 113 mm).

MARGINS OF SAFETY SUMMARY

Case	σ_{max}	$\mathrm{MoS}_{\mathrm{yld}}$	$\mathrm{MoS}_{\mathrm{ult}}$	Location
QSL-000	70.98	+3.90	+3.65	Rail fillet
QSL-015	83.93	+3.15	+2.93	Rail fillet
QSL-045	93.29	+2.73	+2.54	Rail fillet
QSL-090	102.3	+2.40	+2.23	Rail fillet
QSL-135	110.9	+2.14	+1.98	Rail fillet
QSL-180	**138.8**	**+1.51**	**+1.38**	**Rail fillet**
QSL-225	103.2	+2.37	+2.20	Rail fillet
QSL-270	95.40	+2.65	+2.46	Rail fillet
QSL-315	105.1	+2.31	+2.14	Rail fillet

σ_{max} in MPa, rail fillet (Al 7050-T73510: σ_y=435, σ_u=495 MPa). MoS_{yld} = $(\sigma_y/(1.25\,\sigma_{max})) - 1$; MoS_{ult} = $(\sigma_u/(1.5\,\sigma_{max})) - 1$. 15 of 24 angles pending.

STRUCTURAL RESULTS

Check	Result	
$\mathrm{FS}_{\mathrm{yield}}$ (QSL-180)	2.51 > 1.25 req.	DONE
$\mathrm{FS}_{\mathrm{ult}}$ (QSL-180)	2.38 > 1.50 req.	DONE
f_1 (1st eigenfreq.)	287 Hz ≫ 40 Hz	DONE
Max displacement (PLD)	TBD	WIP
Mesh convergence	TBD	WIP
QSL-θ runs complete	9/24	WIP
Norms: ECSS-E-ST-32C · RPUG v10		

DESIGN IMPACT

QSL-180 governs with σ_{max} = 138.8 MPa at the rail fillet under the +10 g axial / −17 g X rideshare combination. All 9 angles run to date show positive margins; lowest $\mathrm{MoS}_{\mathrm{ult}}$ = +1.38 (QSL-180), roughly a fifth the margin of the earlier ECSS-E-ST-32C-only case set, reflecting the harsher RPUG v10 rideshare environment. No design modifications required at this stage.

NEXT STEPS

- Complete remaining 15 QSL-θ angles (30°–345° off-axis, 15° increment)
- Mesh convergence study (h-refinement) at the rail fillet
- Extract max displacement (URES) at the PLD mount for the governing case

STRUCTURAL VERDICT — PASS (9/24 cases)

All 9 QSL-θ rideshare cases run to date satisfy ECSS-E-ST-32C.

$\mathrm{FS}_{\mathrm{yield,min}} = 2.51 > 1.25$ | $\mathrm{FS}_{\mathrm{ult,min}} = 2.38 > 1.50$

Remaining 15 angles in progress; verdict final on completion.